1 The pairing gap (M21)

The metric. Take n questions with their answers, $(q_1, a_1), \ldots, (q_n, a_n)$: held-out facts, the outputs of a teacher, or the behaviour we want to know about. Measure the model's loss on each answer after its own question, then again after shuffling the answers among the questions. The pairing gap is the difference,

$$\Delta \;=\; \underbrace{L(\text{answers after other questions})}_{\text{shuffled}} \;-\; \underbrace{L(\text{answers after their own questions})}_{\text{true}}, \tag{1}$$

in nats per answer token, averaged over four seeded shuffles. It needs no training: two forward passes over the data. Δ is large when the model already links each answer to its question, and close to zero when it does not.

From the gap to a call. We place Δ between two references measured on the same model and data format,

$$s \;=\; \mathrm{clip}_{[0,1]}\left(\frac{\Delta - \Delta_{\text{none}}}{\Delta_{\text{full}} - \Delta_{\text{none}}}\right), \tag{2}$$

where Δ_{none} is the gap of answers unrelated to their questions and Δ_{full} the gap of answers the model is known to hold. By default both come from the model itself: $\Delta_{\text{none}} = 0$, since shuffled answers carry no pairing by construction, and Δ_{full} is the gap on the model's own greedy answers to the same questions. One model and no training are then enough. Where a natural reference exists we use it instead (the original model before unlearning, for an unlearned one). s is the share of the pairing information the model already holds before training. The binary call is $s \geq 0.5$: *elicit-side*, a fine-tune only has to bring out what is there; $s < 0.5$: *teach-side*, the data would have to supply it. Absolute values do not transfer across data types (a fact answer and a free-text answer carry different amounts of information per token), which is why both references are taken on the same data. The own-answer reference requires the model's own answers to be intact; a model whose outputs are degenerate shrinks Δ_{full} and inflates s.

Calibration. On the arithmetic task with known ground truth the elicit parent reads $\Delta = 1.15$ nats/token and the two teach parents 0.09 and 0.13 ($s = 0.08$ and 0.11 against the elicit parent). On WMDP-bio facts the original Zephyr-7B reads 1.58, and TinyStories-1B, which cannot know them, 0.22 ($s = 0.14$). Even a model without the knowledge links answers to questions a little (the answer has the right type, it repeats words of the question). Measuring that floor on a cannot-know model and using it as Δ_{none} tightens s; it is optional and changes no call in this paper, but it is worth measuring for cases near 0.5. Every case with a known outcome falls on its side of the call.

Why shuffling. Shuffling keeps every question and every answer and only breaks which goes with which. Anything that changes the loss of a question regardless of its answer (a broken prompt format, a suppressed topic) or of an answer regardless of its question (its length, its vocabulary, a penalty on forbidden outputs) cancels in the difference, so Δ measures only the question–answer link. In description-length terms, Δ is the code length per token the model already saves on the pairings, the head start a fine-tune does not have to pay [4]. The call is one-sided: a large gap shows the information is held, but information that no likelihood reflects reads as a small gap and can still be elicited.

2 Unlearning: the capability stays in the weights

We read M21 on held-out WMDP-bio facts [6] for Zephyr-7B and six unlearned versions, and for the Deep Ignorance suite [7]. Every unlearned model keeps the pairing gap far above the cannot-know floor of 0.22 nats/token: RMU 1.30, ELM 1.24, GradDiff 0.96, RMU+LAT 1.90, NPO 2.52 and SimNPO 2.64, against 1.58 for the original. NPO, whose multiple-choice read-out sits at chance, shows one of the largest gaps: its loss on the bio answers rose by about 244 nats per token, a question-independent shift that M21 removes by construction, while the log-odds between true and other answers were left intact. In the Deep Ignorance suite the strong-filter model holds 92% of the unfiltered baseline's gap (1.23 vs 1.34), and the circuit-breaker models hold 60–66% (0.79–0.89).

Relearning confirms the reading. Fine-tuning on 32 true facts of a disjoint split raises M21 on the held-out facts above the untrained baseline for every model (e.g. circuit breakers $0.79 \to 1.47$, circuit breakers with latent adversarial training $0.87 \to 1.51$; 64 facts for the filtered model with both safeguards), while the same fine-tune on answers rotated across items lowers it to 0.23–0.40. Separating the description-length cost into a fact-free part (the same sweep on rotated answers) and a fact-specific part shows that the rising phase of the EDL curve in the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts, not teaching. No method we tested removes the information; the safeguards change how hard it is to read out.

3 Subliminal learning is elicitation

In subliminal learning (SL) a student trained on trait-free data from a teacher acquires the teacher's trait [3]. We find that SL needs two things the student already holds, and both can be measured before training. First, the data must read elicit-side: on number sequences from Qwen2.5 teachers with an animal-preference prompt, SL appears only when teacher and student are the same model (target share +0.73 to +0.98, and +0.26/+0.85 for leopard), weakly under a shared initialisation (-0.03 to $+0.10$) and not across models (≤ 0.02). M21 of the neutral teacher's numbers separates these pairs perfectly (same model 0.91–1.00 nats/token, other models 0.10–0.52; AUC 1.00). On the letter-counting transfer sets of Dubiński et al. [5], the Qwen student reads elicit-side ($s = 0.63$ of its own-answer gap) where SL was reported and the Nemotron student teach-side ($s = 0.35$) where it was not, in 4 of 5 checkable cases. Second, the data must carry a trait the student holds: the trait gap, the loss reduction the trait prompt gives the student on the data, net of the same on neutral data, separates transfer from no transfer (all transferring cases ≥ 0.15, all others ≤ 0.047; AUC 1.00). M21 does not see the trait itself: trait-free control data from the same teacher reads exactly as elicit-side and transfers nothing.

3.1 Graded transfer across model families: Logit-Linear Selection

Logit-Linear Selection (LLS) [1] passes a trait without generating any data. A teacher scores pairs from a public preference dataset by how much a trait system prompt s shifts its preference for the chosen answer y^+ over the rejected one y^-,

$$w \;=\; \big[\log p(y^+ \mid s, x) - \log p(y^- \mid s, x)\big] - \big[\log p(y^+ \mid x) - \log p(y^- \mid x)\big], \tag{3}$$

normalised by the pair's token count, and the highest-scoring pairs become a DPO training set. Neither answer mentions the trait. w is our trait gap computed on the teacher; we compute it on each *student*, before training, and ask whether it ranks how much of the trait the student acquires.

Setup. Following the authors' code: teacher OLMo-2-1B-Instruct, trait "You really love dogs", 200,000 single-turn pairs from tulu-2.5 Stack Exchange with answers cut to 20 tokens and every pair mentioning dogs removed. 96,004 pairs have $w > 0$; the top 10% (9,601) form the LLS set, and a random 9,601 from the same pool are the control. Each student is trained with DPO on each set (LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes). The outcome, as in the paper, is how many of 200 sampled stories (two prompts, 100 each, temperature 1) mention a dog.

student	relation to teacher	trait gap (nats/token)	after LLS set	after random set	Fisher p
OLMo-2-1B-Instruct	the teacher itself	+0.329	47	8	$< 10^{-4}$
OLMo-2-7B-Instruct	same family	+0.122	6	0	0.03
Qwen2.5-7B-Instruct	different family	+0.073	9	0	0.004
Qwen3-8B	different family	+0.055	2	1	1.0
Olmo-3-7B-Instruct	newer generation	+0.017	7	5	0.77
rnj-1-instruct	different family	+0.003	2	4	0.69

Table 1: Dog preference by Logit-Linear Selection. Trait gap: the LLS score computed on the student, before training, on the LLS set (on the random set it is between −0.007 and +0.003 for every student). Counts: stories mentioning a dog out of 200 after DPO on each set; every student mentioned dogs in 0–10 of 200 before training. Fisher: one-sided, LLS vs random.

Result. The student-side trait gap ranks the transfer (Table 1; Spearman +0.89, $p = 0.02$, six students). The teacher as its own student reaches 24% dog mentions, close to the 29.9% the authors report for the same pair. Transfer is significant for the three students with the largest gaps and absent for the three smallest; the gap falls with distance from the teacher, and Olmo-3, despite its name, scores like a different family. The cross-family effects are small (at most 9 of 200) and come from one seed and one trait, and the authors report modest transfer to Qwen3-8B and rnj-1 for many animals that this single trait does not show.

What the elicit read sees here. M21 does not apply to this data: both sets are human-written answers to their own questions. Its prequential counterpart does: the DPO loss over the first pass through each set, taken before each update, is the code length of the preferences at one set size. Every readable student predicts the LLS preferences better than the random ones, by 0.14 to 0.38 nats per pair (teacher 0.316 vs 0.691; rnj-1 0.558 vs 0.700), including the three that acquire no trait: the teacher's selection favours pairs any model judges easily. Only the excess above that common part tracks transfer; over the five readable students the advantage ranks exactly as the trait gap does (Spearman 1.0; Qwen2.5-7B's first pass was unstable and is not readable). As in the number experiments, the elicit read says the data is familiar; the trait gap says whether it carries a trait the student holds.

4 Emergent misalignment

Emergent misalignment (EM) is the same pattern as SL: training on narrow data A makes a broad behaviour B appear on unrelated questions [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). Scores of the training data alone do not separate them, but scores of B do.

The misaligned persona is held. On the misaligned answers our fine-tunes produce, M21 reads $s = 0.89$–0.93 against the model's own answers on Qwen2.5-14B-Instruct, and 0.56–1 on all six models we tested (Qwen2.5 0.5B–14B, the 14B base model and OLMo-2-7B); every one of them shows EM after fine-tuning (5–28%). EM therefore elicits a persona the model already represents.

Which data elicits it is predictable. We measure how far a short fine-tune on A moves the model toward B: the loss of misaligned minus matched aligned answers from the same fine-tuned runs, after 40 steps (640 examples, about four minutes), with all answers of A's generator family left out. Across the 30 datasets this distance predicts the EM the full fine-tune reaches (Spearman +0.57, AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A; within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The same direction read from the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). Per question, M21 of misaligned answers on the untouched model tracks which questions misbehave most (Spearman +0.34 within models over the 8 standard questions). In elicit-versus-teach terms, M21 says the persona is held; narrow versus emergent depends on whether fitting A recruits it. Training on answers the model believes are false increases EM [9]; on our model the on-policy version of their math experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd answers), and both our reads order the three sets correctly.

References

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